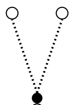
Renormalized equation

$$(\partial_t - \Delta + 1)\, u = u_x^2 g(u) + \xi f(u)$$

with $\xi \in C^{-1-\kappa}$, parabolic scaling $\mathfrak{s} = (2, 1)$, spatial dimension $d = 1$.

Divergent trees

τ	$\lvert\tau\rvert$	$S(\tau)$	k_τ	$F(\tau^*)$
	$-1-\kappa$	1	k_0	$f(u)$
	-2κ	2	k_3	$2f^2(u)g(u)$
	-2κ	1	k_4	$f(u)f'(u)$
	$-\kappa$	1	k_1	$u_x f'(u)$
	$-\kappa$	1	k_2	$2u_x f(u)g(u)$

Legend: ○ = noise, ● = integration node, solid edge = $\mathcal{I}$, dotted edge = $\partial\mathcal{I}$ (derivative).

Renormalized family

$$\begin{aligned}(\partial_t - \Delta + 1)\, u = u_x^2 g(u) &+ \xi f(u) \\ &+ k_0 f(u) \\ &+ k_3 f^2(u)g(u) \\ &+ k_4 f(u)f'(u) \\ &+ k_1 u_x f'(u) \\ &+ 2k_2 u_x f(u)g(u)\end{aligned}$$

Pass `canonical=True` *for the canonical (BPHZ) renormalization $k_\tau = h(S'\_\tau)$ of a centered Gaussian noise (parity-reduced; many constants vanish). Numeric $h(\sigma)$ values need a noise law (Phase 4).*